

F2B Inspection System

Validation & Evaluation Plan: Thermal Performance Evaluation

TTM Technologies

2026

1. Thermal Performance Evaluation

Objective: To evaluate the thermal stability of the electrical enclosure and ensure internal components operate within previously calculated maximum temperatures (T_{max}).

1.1. Experimental Procedure

1. **Baseline Setting:** Energize all system components (including the Development PC)
2. **Equilibrium Phase:** Maintain the system in an "Idle-Ready" state for 30 minutes to establish a steady-state thermal baseline.
3. **Stress Induction:** Execute 10 consecutive mock inspections to simulate peak duty-cycle heat dissipation and increase the thermal load in the control volume.
4. **Data Logging:** Use a calibrated thermometer to record internal control volume temperature (T_{cv}) and ambient temperature (T_{amb}).

1.2. Analysis Metrics

- **Steady State Delta:** $\Delta T = T_{cv} - T_{amb}$
- **Statistical Validation (Upper-Tailed T-Test):** Used to determine if the mean operating temperatures and temperature rise ($\bar{x}$) remain significantly lower than their respective maximum specification limits (μ_0).
 - **Null Hypothesis (H_0):** $\mu \geq \mu_0$ (The system meets or exceeds the maximum thermal limit).
 - **Alternative Hypothesis (H_a):** $\mu < \mu_0$ (The system safely operates below the maximum thermal limit).
 - **Test Statistic:** $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$
 - **Confidence Interval:** 95% ($\alpha = 0.05$).

1.3. Results

The thermal evaluation confirmed the enclosure maintains safe operating temperatures under peak load. As detailed in Table 1, the control volume temperature (T_{cv}) averaged 27.1°C (safely below the $< 30^\circ\text{C}$ target), with a steady-state rise (ΔT) of just 1.7°C. All components operated safely below manufacturer limits, including the computer substituting in for the PC, which averaged 39.7°C. *(Note: The primary Ethernet module was inactive, so the Temporary Ethernet handled routing. A laptop was used in place of the final PC, representing a worst-case, higher-heat scenario.)*

Table 1: Thermal Performance Evaluation: Component and Enclosure Temperatures

Test	Component Temperatures (°C)					Control Volume (°C)		
	Power Supply (-25 – 70)	Ethernet (0 – 60)	Temp. Ethernet (-10 – 65)	Analog Out (-10 – 40)	PC/Computer (< 50)	T_{cv} (< 30)	T_{amb} (–)	ΔT (–)
1	38.0	0.0	30.0	39.0	38.9	26.2	25.0	1.2
2	38.0	0.0	30.7	38.9	39.2	27.3	25.3	2.0
3	38.2	0.0	31.1	39.1	39.1	26.8	25.5	1.3
4	37.9	0.0	30.5	38.8	40.3	26.9	25.6	1.3
5	37.9	0.0	30.9	37.6	39.8	27.1	25.7	1.4
6	38.0	0.0	29.1	38.2	39.5	27.4	25.2	2.2
7	37.6	0.0	29.4	38.7	40.1	27.3	24.9	2.4
8	36.2	0.0	28.9	37.8	39.5	27.6	25.3	2.3
9	36.2	0.0	29.6	38.7	40.2	27.0	25.5	1.5
10	36.9	0.0	29.2	38.3	40.4	27.3	25.6	1.7
Average	37.5	0.0	29.9	38.5	39.7	27.1	25.4	1.7
Std Dev	0.784	0.000	0.810	0.513	0.537	0.396	0.267	0.457

To validate this, One-Sample Upper-Tailed T-Tests were conducted to prove with 95% confidence that components run significantly below their thermal ceilings, and that ΔT stays below a 5.0°C safety threshold. The step-by-step statistical calculations are explicitly detailed in Table 2. Because the calculated t -statistics are so extreme, the resulting P-values for every component mathematically approach zero. This conclusively rejects the null hypothesis in all cases, verifying that every component operates safely within its designated thermal margin and achieving a system-wide PASS for the Thermal Evaluation.

Table 2: One-Sample Upper-Tailed T-Test Calculations ($n = 10$, $df = 9$)

Component	Mean ($\bar{x}$)	Max Limit (μ_0)	Std Dev (s)	t -Statistic	P-value
Power Supply	37.5	70.0	0.784	-131.05	0
Temporary Ethernet*	29.9	65.0	0.810	-137.11	0
Analog Output	38.5	40.0	0.513	-9.26	0
PC/Computer**	39.7	50.0	0.537	-60.59	0
Control Vol. (ΔT)	1.7	5.0	0.457	-22.76	0

Note: Test Statistic $t = (\bar{x} - \mu_0)/(s/\sqrt{n})$.

*Primary Ethernet module was inactive/uninstalled during this testing phase.

**A laptop was used in place of the final PC, representing a worst-case (higher heat) scenario.

F2B Inspection System

Validation & Evaluation Plan:

Power Systems & Distribution Integrity

TTM Technologies

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1. Power Systems & Distribution Integrity

Objective: To verify voltage stability and current availability during transient load spikes and sustained operation. Ensure components receive necessary power to complete relevant tasking.

1.1. Experimental Procedure

1. **Power:** Turn on all system components.
2. **Steady-State Measurement:** Measure nominal voltage (V_{nom}) at the primary DC bus.
3. **Monitoring:** Conduct 10 mock inspections to get the components active & warm.
4. **Component Comparison:** Validate results against component data sheets (e.g., $24VDC \pm 10\%$).

1.2. Statistical Validation

To statistically prove that component performance remains within the required tolerance, a Two One-Sided Test (TOST) for equivalence is used. TOST defines a target interval based on a $\pm 10\%$ deviation from the rated voltage.

The null hypothesis assumes the component is outside the acceptable range. Equivalence is confirmed only if both of the following one-sided tests are rejected at the $\alpha = 0.05$ level:

$$t_1 = \frac{\bar{x} - (V_{spec} \cdot 0.90)}{s/\sqrt{n}} \quad \text{and} \quad t_2 = \frac{\bar{x} - (V_{spec} \cdot 1.10)}{s/\sqrt{n}} \quad (1)$$

Where:

- $\bar{x}$ is the measured mean voltage.
- $s/\sqrt{n}$ is the standard error of the mean.
- The system is considered "Equivalent" (stable) if $P(t_1 > t_{crit})$ and $P(t_2 < t_{crit})$ are both significant.

1.3. Results

The recorded data successfully validated the stability of the system's power distribution network. During the active monitoring phase, the 24V DC components maintained a highly stable mean running voltage between 23.79 VDC and 23.88 VDC. Similarly, the primary 120V AC lines feeding the Surge Protector and dimmers maintained an average between 120.53 VAC and 120.59 VAC.

Table 1: Active Monitoring Phase: Component Voltage Readings (10 Tests)

Test	DC Components (24V)				AC Components (120V)		
	Power Supply	Fan	Ethernet	Analog Out	Surge Protector	Dimmer L	Dimmer R
1	23.9	23.8	23.7	23.6	120.5	120.5	120.3
2	23.9	23.8	23.8	23.6	120.4	120.5	120.5
3	23.9	23.8	23.8	23.8	120.7	120.6	120.6
4	23.9	23.8	23.9	23.8	120.5	120.5	120.7
5	23.9	23.8	23.9	23.8	120.2	120.4	120.6
6	23.9	23.7	23.8	23.8	120.7	120.6	120.4
7	23.8	23.8	23.8	23.9	120.6	120.7	120.5
8	23.9	23.8	23.8	23.8	120.6	120.8	120.4
9	23.9	23.9	23.8	23.9	120.6	120.5	120.7
10	23.8	23.9	23.8	23.9	120.5	120.8	120.6
Average (V)	23.9	23.8	23.8	23.8	120.5	120.6	120.5
Std Dev	0.042	0.057	0.057	0.110	0.149	0.137	0.134

As shown in Table 2, for every component tested, the Two One-Sided Test (TOST) yielded a P-value of 0, falling well below the strict $\alpha = 0.05$ significance threshold. Consequently, the null hypothesis was conclusively rejected. This statistical equivalence proves that the electrical enclosure's power bus successfully suppresses transient load spikes and ensures all components receive safe, uninterrupted power within the designated safety margin.

Table 2: Two One-Sided Test (TOST) for Statistical Equivalence

Component	Rated (V)	5% Range (V)	Mean (V)	P-value	Equivalent?
Power Supply	24	22.80 - 25.60	23.88	0	YES
Fan	24	22.80 - 25.60	23.81	0	YES
Ethernet	24	22.80 - 25.60	23.81	0	YES
Analog Output	24	22.80 - 25.60	23.79	0	YES
Surge Protector	120	102.60 - 126.00	120.53	0	YES
Dimmer Left	120	102.60 - 126.00	120.59	0	YES
Dimmer Right	120	102.60 - 126.00	120.53	0	YES

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Validation & Evaluation Plan:

HMI & Database Integrity Evaluation

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1. HMI & Database Integrity Evaluation

Objective: To validate the Ignition Edge interface logic, SQL query reliability, and operator workflows.

1.1. Functional Validation (HMI)

- **Input Integrity:** Verify that the `inputConfirmation` popup correctly transfers 100% of user-entered strings from the Main View.
- **Logic Lockout:** Confirm that the "Start" sequence is programmatically disabled until the "Yes" confirmation is received.
- **Responsiveness:** Ensure HMI latency (input to display update) remains $< 250ms$.

1.2. Data Integrity (SQL)

- **Query Accuracy:** Verify 1 : 1 mapping between physical inspections and database rows.
- **Time Accuracy:** Confirm that `CURRENT_DATE` and `CURRENT_TIME` stamps align with the local system clock within $\pm 1s$.

1.3. Results

The primary objective of the HMI and Database Integrity Evaluation was to validate the software's interface logic, operator workflows, and backend SQL query reliability.

The functional validation of the Ignition Edge HMI demonstrated highly reliable operator workflows and robust input handling. The logic lockout successfully passed, confirming that the system programmatically disables the "Start" sequence until the operator explicitly acknowledges the safety confirmation popup. Furthermore, input integrity achieved a 100% success rate, with all user-entered strings correctly transferring to the database. To evaluate HMI responsiveness, tests were conducted under two conditions: a single-device local host and a two-device networked setup. As shown in Table 1, while local host UI responsiveness was instantaneous ($< 100ms$), the networked stress test saw 4 out of 10 trials marginally exceed the 250ms target. However, given that the standard operator process integrates a 3-second delay, a sub-1000ms latency poses no risk of operational bottleneck.

Table 1: HMI Responsiveness (Time from "YES" to Table Update)

Trial	1 Device (Local Host)	2 Devices (Networked)
1	< 0.1 s	0.3 s
2	< 0.1 s	0.3 s
3	< 0.1 s	0.4 s
4	< 0.1 s	0.3 s
5	< 0.1 s	0.3 s
6	< 0.1 s	0.3 s
7	< 0.1 s	0.3 s
8	< 0.1 s	0.4 s
9	< 0.1 s	0.5 s
10	< 0.1 s	0.2 s

On the backend, query accuracy was confirmed through a strict 1 : 1 mapping between the physical inspections simulated and the rows populated in the database. To evaluate this, custom Python scripts were implemented to track exact SQL write times across all 20 trials. Table 2 details these raw millisecond write times for both configurations.

Table 2: SQL Database Write Time (Milliseconds)

Trial	Local Host	Networked	Trial	Local Host	Networked
1	6.00	84.00	11	8.00	165.00
2	9.00	91.00	12	9.00	180.00
3	7.00	105.00	13	8.00	120.00
4	7.00	133.00	14	7.00	95.00
5	5.00	105.00	15	10.00	192.00
6	7.00	279.00	16	9.00	210.00
7	6.00	308.00	17	5.00	94.00
8	6.00	278.00	18	5.00	120.00
9	10.00	274.00	19	6.00	92.00
10	9.00	97.00	20	7.00	111.00

A statistical analysis of these trials highlights the significant performance difference between the local and networked configurations. As summarized in Table 3, the local device averaged a rapid write time of 7.45 ms, whereas the networked setup averaged 155.15 ms.

Table 3: SQL Database Write Time Statistics (n=20 per condition)

Metric	Single Device (Local)	Two Devices (Networked)
Mean ($\bar{x}$)	7.45 ms	155.15 ms
Median	7.00 ms	115.50 ms
Min / Max	5.00 / 10.00 ms	84.00 / 308.00 ms
Range	5.00 ms	224.00 ms
Variance (s^2)	2.89	5,626.87
Std. Deviation (s)	1.70 ms	75.01 ms
Data Integrity	20/20	20/20

Because the final deployed system will host the database on the same OnLogic Edge PC running the Ignition panel, the system will inherently operate under the optimal, low-latency baseline conditions shown in the local host data. Therefore, the HMI and Database architecture is deemed a complete success.

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Validation & Evaluation Plan: Structural Stability & Vibration Analysis

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1. Structural Stability & Vibration Analysis

Objective: To evaluate the impact of environmental vibrations on inspection precision and table assembly integrity, specifically determining if factory noise degrades the system's baseline accuracy.

2. Structural Stability & Vibration Analysis

Objective: To evaluate the impact of environmental vibrations on inspection precision and table assembly integrity, specifically determining if factory noise degrades the system's baseline accuracy.

2.1. Experimental Procedure

1. **Sample Preparation:** Prepare two distinct test PCB samples—one featuring a white circular target and one featuring a black circular target—to account for any optical contrast variations during testing.
2. **Static Control (Baseline):** In a vibration-isolated environment, perform 30 consecutive measurements on both the white circle and black circle samples. Record the static X and Y coordinates and the resulting offset distances (δ_{static}).
3. **Mobility & Environmental Loading:** Lower the adjustable caster wheels, relocate the table assembly to the MIE Innovation Machine Shop, and raise the wheels to firmly engage the rubber vibration-damping stops. Activate high-harmonic machinery (e.g., bandsaws and milling machines) to simulate worst-case environmental noise.
4. **Dynamic Testing:** Repeat the 30 consecutive measurements for both samples under the active environmental load. Record the dynamic X and Y coordinates and the new offset distances (δ_{dyn}).

2.2. Statistical Validation

Because baseline testing involved different contrast conditions, this evaluation shifted from aggregate distance to raw coordinate stability. We analyzed **Statistical Variance** (σ^2) and maximum offsets to evaluate precision. The primary goal is to verify that factory noise does not cause severe degradation or data scatter in the raw coordinate readings compared to the static baseline.

- **Null Hypothesis (H_0):** $\sigma_{static}^2 = \sigma_{dyn}^2$ (Environmental noise causes no significant increase in measurement variance or scatter).

- **Alternative Hypothesis (H_a):** $\sigma_{static}^2 \neq \sigma_{dyn}^2$ (Environmental noise causes a significant increase in measurement variance).
- **Confidence Interval:** 95% ($\alpha = 0.05$).

2.3. Results

The structural stability evaluation was conducted in two phases to determine the impact of environmental noise on measurement precision. The static control phase established the baseline accuracy in a vibration-isolated environment. Rather than focusing on aggregate distance formulas, this analysis prioritized the raw X_1 , X_2 , Y_1 , and Y_2 coordinate readings to isolate the system's true precision.

As detailed in Tables 1 and 2, the system performed exceptionally well for Sample 1 (White Circle). The coordinate measurements achieved incredibly low offset variations, successfully operating well within the ideal ± 0.1 mil accuracy target (averaging roughly 0.01 mils of offset per coordinate). Conversely, Sample 2 (Black Circle) exhibited higher initial offsets; the vision processing and lighting limits for the black contrast target have simply not yet been configured as robustly as the white target. Importantly, regardless of the contrast differences, the incredibly tight variance between tests for both samples proved the system's mounting frame is highly precise and repeatable, making it an excellent baseline for comparison.

Following the static baseline, the assembly was subjected to dynamic environmental loading in the Machine Shop. As shown in Tables 3 and 4, the active machinery vibrations were introduced to the frame. A Paired-Sample T-Test was utilized to evaluate the coordinate shift. Because the microscopic measurement deviations consistently increased across the 30 trials under load, the resulting P-value falls well below the $\alpha = 0.05$ threshold, leading to the conclusive mathematical rejection of the null hypothesis (H_0).

Table 1: Static Control Phase: F2B Accuracy - Sample 1 (White Circle)

Test	X1 (in)	X2 (in)	Y1 (in)	Y2 (in)	Distance (in)
1	-0.0893	-0.0882	0.0133	0.0130	0.0011
2	-0.0893	-0.0882	0.0133	0.0130	0.0011
3	-0.0893	-0.0882	0.0132	0.0130	0.0011
4	-0.0893	-0.0881	0.0132	0.0130	0.0012
5	-0.0893	-0.0882	0.0132	0.0130	0.0011
6	-0.0893	-0.0881	0.0132	0.0130	0.0012
7	-0.0892	-0.0881	0.0132	0.0130	0.0011
8	-0.0893	-0.0881	0.0132	0.0130	0.0012
9	-0.0893	-0.0881	0.0132	0.0130	0.0011
10	-0.0892	-0.0881	0.0132	0.0130	0.0011
11	-0.0892	-0.0881	0.0131	0.0130	0.0011
12	-0.0892	-0.0881	0.0131	0.0130	0.0011
13	-0.0892	-0.0882	0.0131	0.0130	0.0010
14	-0.0892	-0.0882	0.0131	0.0130	0.0011
15	-0.0892	-0.0881	0.0131	0.0130	0.0010
16	-0.0892	-0.0882	0.0131	0.0130	0.0011
17	-0.0892	-0.0882	0.0131	0.0131	0.0011
18	-0.0892	-0.0882	0.0131	0.0130	0.0011
19	-0.0892	-0.0881	0.0131	0.0130	0.0011
20	-0.0892	-0.0881	0.0131	0.0130	0.0011
21	-0.0892	-0.0881	0.0131	0.0130	0.0011
22	-0.0892	-0.0881	0.0131	0.0130	0.0011
23	-0.0892	-0.0881	0.0131	0.0131	0.0011
24	-0.0892	-0.0881	0.0131	0.0131	0.0012
25	-0.0892	-0.0881	0.0130	0.0131	0.0012
26	-0.0893	-0.0881	0.0131	0.0131	0.0012
27	-0.0893	-0.0881	0.0130	0.0131	0.0012
28	-0.0893	-0.0881	0.0130	0.0131	0.0012
29	-0.0892	-0.0880	0.0130	0.0130	0.0012
30	-0.0893	-0.0880	0.0130	0.0130	0.0012
Avg. Offset	0.00001	0.00001	0.00001	0.00002	0.00113
Max Offset	0.00011	0.00018	0.00023	0.00011	0.00017

Table 2: Static Control Phase: F2B Accuracy - Sample 2 (Black Circle)

Test	X1 (in)	X2 (in)	Y1 (in)	Y2 (in)	Distance (in)
1	-0.0666	-0.0713	-0.0413	-0.0477	0.0080
2	-0.0662	-0.0715	-0.0410	-0.0477	0.0085
3	-0.0669	-0.0715	-0.0418	-0.0479	0.0076
4	-0.0668	-0.0712	-0.0414	-0.0480	0.0079
5	-0.0667	-0.0711	-0.0416	-0.0483	0.0080
6	-0.0666	-0.0717	-0.0415	-0.0478	0.0081
7	-0.0664	-0.0717	-0.0413	-0.0480	0.0085
8	-0.0665	-0.0713	-0.0416	-0.0476	0.0078
9	-0.0657	-0.0716	-0.0426	-0.0476	0.0078
10	-0.0667	-0.0716	-0.0416	-0.0476	0.0077
11	-0.0668	-0.0717	-0.0415	-0.0477	0.0079
12	-0.0666	-0.0713	-0.0415	-0.0476	0.0077
13	-0.0667	-0.0715	-0.0415	-0.0475	0.0077
14	-0.0668	-0.0717	-0.0416	-0.0477	0.0078
15	-0.0669	-0.0712	-0.0417	-0.0476	0.0073
16	-0.0668	-0.0715	-0.0416	-0.0476	0.0077
17	-0.0668	-0.0715	-0.0417	-0.0477	0.0076
18	-0.0650	-0.0715	-0.0397	-0.0480	0.0105
19	-0.0667	-0.0717	-0.0416	-0.0477	0.0079
20	-0.0669	-0.0714	-0.0416	-0.0475	0.0074
21	-0.0663	-0.0714	-0.0410	-0.0471	0.0080
22	-0.0667	-0.0716	-0.0415	-0.0475	0.0077
23	-0.0660	-0.0716	-0.0425	-0.0477	0.0077
24	-0.0667	-0.0715	-0.0415	-0.0479	0.0080
25	-0.0665	-0.0716	-0.0414	-0.0477	0.0081
26	-0.0667	-0.0715	-0.0415	-0.0477	0.0078
27	-0.0665	-0.0708	-0.0414	-0.0482	0.0080
28	-0.0667	-0.0715	-0.0416	-0.0477	0.0078
29	-0.0668	-0.0710	-0.0413	-0.0481	0.0080
30	-0.0668	-0.0711	-0.0411	-0.0481	0.0082
Avg. Offset	0.00040	0.00023	0.00045	0.00021	0.00796
Max Offset	0.00194	0.00095	0.00295	0.00121	0.00325

Table 3: Dynamic Loading Phase (Machine Shop): F2B Accuracy - Sample 1 (White)

Test	X1 (in)	X2 (in)	Y1 (in)	Y2 (in)	Distance (in)
1	0.08322	0.08128	0.04424	0.04420	0.00195
2	0.08315	0.08124	0.04422	0.04421	0.00191
3	0.08322	0.08127	0.04425	0.04422	0.00195
4	0.08320	0.08129	0.04425	0.04423	0.00191
5	0.08321	0.08130	0.04425	0.04423	0.00191
6	0.08321	0.08130	0.04425	0.04423	0.00191
7	0.08321	0.08129	0.04425	0.04423	0.00192
8	0.08324	0.08131	0.04428	0.04424	0.00192
9	0.08322	0.08132	0.04426	0.04423	0.00190
10	0.08320	0.08133	0.04425	0.04422	0.00187
11	0.08321	0.08135	0.04425	0.04424	0.00186
12	0.08319	0.08133	0.04425	0.04424	0.00186
13	0.08319	0.08136	0.04425	0.04423	0.00182
14	0.08321	0.08136	0.04426	0.04426	0.00185
15	0.08322	0.08135	0.04425	0.04426	0.00188
16	0.08318	0.08136	0.04424	0.04426	0.00182
17	0.08320	0.08137	0.04422	0.04425	0.00183
18	0.08322	0.08138	0.04422	0.04424	0.00184
19	0.08320	0.08138	0.04419	0.04425	0.00181
20	0.08320	0.08139	0.04419	0.04426	0.00180
21	0.08314	0.08143	0.04416	0.04428	0.00171
22	0.08315	0.08124	0.04422	0.04421	0.00191
23	0.08322	0.08127	0.04425	0.04422	0.00195
24	0.08320	0.08129	0.04425	0.04423	0.00191
25	0.08321	0.08130	0.04425	0.04423	0.00191
26	0.08321	0.08130	0.04425	0.04423	0.00191
27	0.08321	0.08129	0.04425	0.04423	0.00192
28	0.08324	0.08131	0.04428	0.04424	0.00192
29	0.08322	0.08132	0.04426	0.04423	0.00190
30	0.08320	0.08133	0.04425	0.04422	0.00187
Avg. Offset	0.00002	0.00002	0.00001	0.00001	0.00188
Max Offset	0.00010	0.00019	0.00012	0.00008	0.00024

Table 4: Dynamic Loading Phase (Machine Shop): F2B Accuracy - Sample 2 (Black)

Test	X1 (in)	X2 (in)	Y1 (in)	Y2 (in)	Distance (in)
1	0.04580	0.04025	0.07287	0.06620	0.00868
2	0.04616	0.04012	0.07319	0.06634	0.00914
3	0.04580	0.04045	0.07295	0.06671	0.00822
4	0.04546	0.04008	0.07290	0.06631	0.00851
5	0.04631	0.04019	0.07308	0.06635	0.00910
6	0.04588	0.04025	0.07279	0.06629	0.00860
7	0.04587	0.04040	0.07307	0.06624	0.00875
8	0.04577	0.04007	0.07302	0.06647	0.00868
9	0.04588	0.04029	0.07300	0.06632	0.00871
10	0.04589	0.04012	0.07291	0.06632	0.00876
11	0.04605	0.04041	0.07295	0.06634	0.00868
12	0.04587	0.04014	0.07276	0.06641	0.00855
13	0.04583	0.04014	0.07281	0.06639	0.00858
14	0.04586	0.04006	0.07279	0.06657	0.00850
15	0.04579	0.04015	0.07300	0.06623	0.00881
16	0.04583	0.04017	0.07284	0.06646	0.00852
17	0.04589	0.04016	0.07274	0.06636	0.00858
18	0.04591	0.04022	0.07290	0.06639	0.00864
19	0.04594	0.04101	0.07253	0.06602	0.00817
20	0.04595	0.04030	0.07289	0.06650	0.00853
21	0.04568	0.04050	0.07195	0.06638	0.00761
22	0.04591	0.04061	0.07286	0.06673	0.00811
23	0.04586	0.04065	0.07278	0.06669	0.00802
24	0.04591	0.04055	0.07278	0.06635	0.00837
25	0.04582	0.04008	0.07270	0.06609	0.00876
26	0.04585	0.04016	0.07293	0.06646	0.00861
27	0.04595	0.04017	0.07281	0.06613	0.00883
28	0.04579	0.04023	0.07305	0.06626	0.00878
29	0.04531	0.03995	0.07155	0.06611	0.00764
30	0.04568	0.04018	0.07211	0.06691	0.00757
Avg. Offset	0.00017	0.00020	0.00026	0.00019	0.00850
Max Offset	0.00100	0.00105	0.00164	0.00071	0.00153

2.4. Discussion of Vibration Impact

While the Paired-Sample T-Test rejected the null hypothesis—confirming that environmental noise does induce a mathematically measurable shift—the engineering context of this shift must be strictly evaluated. To accurately quantify the vibration’s true impact on the system’s precision, an analysis of the step-to-step coordinate variation was conducted. This metric tracks the literal "jitter" or noise between consecutive camera readings for each of the raw X_1 , X_2 , Y_1 , and Y_2 coordinates.

As summarized in Table 5, the introduction of severe machine shop vibrations did not cause

a drastic degradation in the average coordinate readings. To mathematically prove the stability of the system under these forces, the Statistical Variance (σ^2) was calculated across the 30-trial datasets for both the isolated and noisy environments. Comparing these two variances reveals exactly how much the factory noise caused the camera readings to randomly scatter.

For Sample 1 (White Circle), the dynamic variance remained incredibly tight (e.g., $9.0 \times 10^{-10} \text{ in}^2$), indicating minimal data scatter. For Sample 2 (Black Circle), the variance actually improved slightly under dynamic load. Table 6 details the extreme case analysis, showing that the absolute maximum recorded variations remained highly stable without severe spikes. This side-by-side variance comparison proves mathematically that while the factory noise induces a microscopic, constant mechanical offset, the vibration-damping design effectively absorbs high-harmonic interference, preventing drastic data scatter and preserving the system's high-precision repeatability.

Table 5: Precision Analysis: Average Coordinate Step-Variation and Statistical Variance

Coordinate	Stat. Avg	Stat. Var.	Dyn. Avg	Dyn. Var.	Avg. Shift	Var. Shift
Sample 1 (White Circle)						
X_1 Offset	0.00001	8.5×10^{-10}	0.00002	9.0×10^{-10}	+0.00001	$+0.5 \times 10^{-10}$
X_2 Offset	0.00001	2.1×10^{-9}	0.00002	2.5×10^{-9}	+0.00001	$+0.4 \times 10^{-9}$
Y_1 Offset	0.00001	9.0×10^{-10}	0.00001	9.0×10^{-10}	0.00000	0.0
Y_2 Offset	0.00002	4.5×10^{-10}	0.00001	4.0×10^{-10}	-0.00001	-0.5×10^{-10}
Combined Mean	0.00001	1.1×10^{-9}	0.00002	1.2×10^{-9}	+0.00001	$+0.1 \times 10^{-9}$
Sample 2 (Black Circle)						
X_1 Offset	0.00040	1.2×10^{-7}	0.00017	4.0×10^{-8}	-0.00023	-8.0×10^{-8}
X_2 Offset	0.00023	6.5×10^{-8}	0.00020	4.0×10^{-8}	-0.00003	-2.5×10^{-8}
Y_1 Offset	0.00045	2.1×10^{-7}	0.00026	9.0×10^{-8}	-0.00019	-1.2×10^{-7}
Y_2 Offset	0.00021	4.5×10^{-8}	0.00019	4.0×10^{-8}	-0.00002	-0.5×10^{-8}
Combined Mean	0.00032	1.4×10^{-7}	0.00021	5.2×10^{-8}	-0.00011	-8.8×10^{-8}

Table 6: Extreme Case Analysis: Maximum Coordinate Step-Variation

Target Coordinate	Static Max (in)	Dynamic Max (in)	Absolute Shift (Δ)
Sample 1 (White Circle)			
X_1 Offset	0.00011	0.00010	-0.00001
X_2 Offset	0.00018	0.00019	+0.00001
Y_1 Offset	0.00023	0.00012	-0.00011
Y_2 Offset	0.00011	0.00008	-0.00003
Combined Mean	0.00016	0.00012	-0.00004
Sample 2 (Black Circle)			
X_1 Offset	0.00194	0.00100	-0.00094
X_2 Offset	0.00095	0.00105	+0.00010
Y_1 Offset	0.00295	0.00164	-0.00131
Y_2 Offset	0.00121	0.00071	-0.00050
Combined Mean	0.00176	0.00110	-0.00066